Demo version of Midterm test on DE
You have 80 minutes

1. Solve first-order DE

a) $x^3y' + x^2y = x^2 - 1,$ b) $xdy - ydx = x\sqrt{x^2 + y^2}\,dx.$

2. Solve linear DE with constant coefficients

a) $y^{(4)} + 2y'' + y = x^2 + 9\sin 2x,$ b) $y''' - y'' + 4y' - 4y = 40\cos^2 x,$ c) $y'' - y' = -\frac{x+1}{x^2}$

3. Solve systems of DE

a) $\begin{cases} \frac{dx}{dt} = 2x + 4y - z, \\ \frac{dy}{dt} = -2x - 7y + 4z, \\ \frac{dz}{dt} = -5x - 10y + 4z \end{cases}$ Eigenvalues $-1, 3i, -3i$

b) $\begin{cases} \frac{dx}{dt} = -x - 2y + 2z, \\ \frac{dy}{dt} = -4x - 2y - 3z, \\ \frac{dz}{dt} = -3x + 3y - 6z \end{cases}$ Multiple eigenvalue -3

4. Given second order differential equation with constant coefficients $y'' + ay' + by = 0$. Let initial values $y(0)^2 + (y'(0))^2 \neq 0$. Find conditions (necessary and sufficient) on the coefficients a, b that guarantee that every solution of this equation in absolute value $|y(x)|$ will monotonically increase starting with some x_0.